

Syllabus: High Availability and Disaster Recovery

Redundancy, failover mechanisms, disaster recovery strategies (backup & restore, pilot light, warm standby), RTO, RPO, Service Level Agreements (SLA).

Redundancy in cloud computing is the practice of duplicating critical system components—such as servers, data, and network paths—to ensure that if one fails, a backup can take over immediately. It is a proactive strategy designed to improve **reliability**, **fault tolerance**, and **high availability**.

Core Types of Redundancy

Data Redundancy: Storing identical copies of data across different disks, servers, or regions to prevent loss from corruption or hardware failure.

Hardware Redundancy: Deploying duplicate physical or virtual assets (servers, power supplies, cooling systems) so operations continue if a specific machine fails.

Network Redundancy: Providing multiple network paths and carriers to the internet to ensure connectivity even if a main line is cut or a provider goes down.

Geographic Redundancy: Replicating data and applications across physically distant data centers (different regions or continents) to protect against localized natural disasters or power outages.

Implementation Strategies

Cloud providers and enterprises use several mechanisms to manage these redundant assets:

Load Balancing: Distributes incoming traffic across multiple redundant servers to prevent any single one from being overwhelmed.

Failover Mechanisms: Automatically redirects workloads from a failed component to a healthy redundant one.

Replication: The process of keeping redundant data synchronized, which can be **synchronous** (immediate update, higher latency) or **asynchronous** (background update, potential data lag).

Comparison of Infrastructure Redundancy Levels

Data centers often categorize their physical redundancy using "N" notation, which indicates the minimum capacity needed to support the load.

Redundancy Level	Description
N	No redundancy; single point of failure.
N+1	One extra component (e.g., 5 UPS for a load needing 4).
2N	Fully mirrored systems with two independent paths.
2N+1	Mirrored systems plus an extra spare for each.

Key Benefits vs. Challenges

Benefits: Ensures business continuity, meets Service Level Agreements (SLAs), and enables disaster recovery.

Challenges: Significant increased costs due to paying for idle or duplicate resources, and higher architectural complexity to manage synchronization and failover

Disaster recovery (DR)

DR in cloud computing involves strategies to back up and restore data, applications, and infrastructure using cloud resources. These strategies are categorized by their balance of cost, complexity, and recovery objectives, specifically Recovery Time Objective (RTO) (maximum downtime) and Recovery Point Objective (RPO) (maximum data loss).

RTO and RPO

RTO (Recovery Time Objective)

RTO is about "Downtime."

It is the maximum amount of time it should take to get your systems back up and running after a disaster occurs.

Question it answers: "How long can we afford for the system to be offline?"

Goal: To restore service as quickly as possible.

Example: If your RTO is 2 hours, your IT team must have the application fully functional within 2 hours of the crash.

RPO (Recovery Point Objective)

RPO is about "Data Loss."

It refers to the maximum amount of data (measured in time) that can be lost from your last backup before it causes serious damage to the business.

Question it answers: "How much data can we afford to lose?"

Goal: To define the frequency of your backups.

Example: If you back up your data every 24 hours and the system crashes right before the next backup, you lose 24 hours of work. If your RPO is 1 hour, you must back up your data at least every hour.

Comparison across Cloud DR Strategies

Strategy	RPO (Data Loss)	RTO (Downtime)
Backup & Restore	High (Hours/Days)	High (Hours/Days)
Pilot Light	Low (Minutes)	Medium (Hours)
Warm Standby	Lower (Minutes)	Low (Minutes)
Active/Active	Near Zero	Near Zero

Key Differences

Focus: RTO focuses on the application and infrastructure (getting them "on"), while RPO focuses on the data (getting it "back").

Cost: Achieving a lower RTO/RPO (near zero) is significantly more expensive because it requires constant synchronization and "live" standby servers.

Cloud providers (such as AWS, Azure, and Google Cloud) typically define four primary DR approaches:

Backup and Restore: The simplest and most cost-effective method where data is backed up to cloud storage (e.g., Amazon S3) and restored only after a disaster. It has the highest RTO as systems must be rebuilt from scratch.

Key Performance Metrics

Because this strategy requires rebuilding infrastructure and transferring data after a disaster, it typically has the longest recovery times.

Recovery Time Objective (RTO): Usually 4 to 24 hours. This represents the time needed to provision new servers and restore data.

Recovery Point Objective (RPO): Usually 1 to 24 hours, depending on backup frequency. It represents the potential data loss since the last backup.

Common Backup Methods

Full Backup: A complete copy of all data; easy to restore but slow and storage-intensive.

Incremental Backup: Only copies data changed since the last backup (of any kind); fastest and most storage-efficient.

Differential Backup: Copies data changed since the last full backup; faster to restore than incremental but grows in size over time.

Snapshots: Point-in-time "images" of a disk or application, often used for rapid recovery of specific virtual machines or databases.

Top Tools and Services (2025-2026)

Category	Top Providers/Tools
Cloud-Native	AWS Backup, Azure Backup, Google Cloud Backup & DR
Enterprise SaaS	Veeam, Druva, Rubrik, Cohesity
Specialized/SMB	Backblaze B2, Acronis Cyber Protect, IDrive

Best Practices for Implementation

Follow the 3-2-1 Rule: Keep 3 copies of data on 2 different media types with 1 copy stored offsite (e.g., in a different cloud region).

Automate Everything: Use Infrastructure as Code (IaC) to automatically redeploy your environment (servers, networks, firewalls) so you don't have to configure them manually during a crisis.

Ensure Immutability: Use features like S3 Object Lock to create backups that cannot be deleted or encrypted by ransomware.

Test Regularly: A backup is only useful if it actually restores. Perform quarterly "restore drills" to verify data integrity and document recovery steps in a Runbook

Pilot Light: Replicates critical data to the cloud while keeping a minimal version of the core infrastructure "switched off" or in standby. In a disaster, these core elements are quickly scaled up to full production capacity.

Key Performance Metrics

Pilot Light offers a balance between the low cost of "Backup and Restore" and the high speed of "Warm Standby".

Recovery Time Objective (RTO): 10s of minutes to a few hours. Time is required to provision and scale up the application tier.

Recovery Point Objective (RPO): Minutes. Data is typically replicated in near real-time to the secondary region.

How it Works

Core Data is "Live": Databases and critical data stores are continuously synchronized (replicated) to the DR region.

Infrastructure is "Idle": Core networking (VPC, subnets) is pre-configured, but application servers (like EC2) are either "switched off" or not deployed at all.

Rapid Activation: In a disaster, the "idle" servers are quickly started or provisioned using pre-built images (e.g., AMIs), and traffic is redirected via DNS.

Common Use Case

An e-commerce site might keep its product database and user records replicating in real-time to a secondary region (Pilot Light). The web servers remain off until an outage is detected, at which point an Auto Scaling Group automatically launches them to restore the site.

Warm Standby: Maintains a scaled-down but fully functional copy of the production environment always running in the cloud. It offers faster recovery than a pilot light because the system is already "on" but at reduced capacity.

Key Performance Metrics

Warm Standby is faster than Pilot Light because the infrastructure is already "on," just not at full strength.

Recovery Time Objective (RTO): Minutes. You only need to scale up the existing instances, not provision new ones from scratch.

Recovery Point Objective (RPO): Near Zero to Minutes. Data is usually replicated synchronously or near-synchronously.

How it Works

Always Running: A minimal version of every tier (Web, App, DB) is active. For example, if production has 20 servers, Warm Standby might have 2.

Live Data: Databases are continuously synced using tools like AWS Aurora Global Database or Azure SQL Failover Groups.

Active Testing: Because the site is "warm," you can send a small percentage of test traffic (e.g., 1%) to it via Route 53 Weighted Routing to ensure it works.

Instant Scaling: During a disaster, your Auto Scaling Group triggers and rapidly expands the 2 servers into 20 to handle the full production load.

Pros and Cons

Pros: Much faster recovery than Pilot Light; allows for continuous "health checks" of the DR site; handles a partial load immediately.

Cons: Higher cost than Pilot Light (you pay for running servers 24/7); more complex to manage than simple backups.

Best For

Business-critical applications where an hour of downtime is unacceptable but the massive cost of a full Active/Active setup isn't justified.

Testing environments that need to double as emergency backup sites.

Comparison: Pilot Light vs. Warm Standby

The two are often confused, but the difference lies in readiness to handle traffic:

Feature	Pilot Light	Warm Standby
Running State	Services are idle/not deployed	Small fleet is always running
Traffic Handling	Cannot handle traffic immediately	Can handle traffic at reduced capacity

Recovery Steps	Start/Deploy + Scale up	Scale up only
Cost	Lower	Higher

Multi-Site (Active/Active): The most expensive and complex strategy where workloads run simultaneously in two or more regions. Traffic is balanced across all sites, providing near-zero RTO and RPO because failover is nearly instantaneous. In a cloud context, Active/Active Multi-Site means your application is running simultaneously in two or more separate locations (Regions or Availability Zones) at all times. Instead of one site sitting idle waiting for a crash, both sites are "live" and actively handling user traffic. AWS and Azure often refer to this as the "zero-downtime" gold standard.

How it Works

Traffic Routing: A global load balancer (like Cloudflare or AWS Route 53) detects where users are and sends them to the "nearest" or "healthiest" site.

Data Synchronization: Databases are kept in sync using bi-directional replication. If a user updates their profile in Site A, that change is immediately mirrored in Site B.

Instant Failover: If Site A experiences a total blackout, the load balancer simply stops sending traffic there. Because Site B is already running at full capacity and has all the current data, users often don't even notice a glitch.

Why use it?

Zero RTO/RPO: Recovery is nearly instantaneous because there is no "startup" time.

Performance: Users get lower latency by connecting to the geographically closest data center.

Resource Efficiency: You aren't paying for "dark" servers that only run during emergencies; you are using 100% of your paid infrastructure.

The Trade-offs

Cost: You are effectively paying double (or more) for compute, storage, and data transfer.

Complexity: Managing "split-brain" scenarios—where two sites disagree on which data is the most recent—requires advanced database engineering (like Amazon Aurora Global Database or Google Cloud Spanner)

Implementation and Best Practices

An effective cloud DR plan requires a structured lifecycle:

Risk Assessment & BIA: Identify potential threats (e.g., cyberattacks, natural disasters) and conduct a Business Impact Analysis (BIA) to prioritize mission-critical systems.

Define Objectives: Set clear RTO and RPO targets for each application tier to guide technology selection.

Automation & Orchestration: Use tools like AWS CloudFormation or Azure Site Recovery to automate infrastructure deployment and failover, reducing human error during a crisis.

Regular Testing: Conduct frequent drills (tabletop exercises, simulations, or full drills) to ensure the plan remains viable as the environment changes.

Multi-Cloud/Geo-Redundancy: Distribute backups across different geographic regions or even different cloud providers (Multi-Cloud DR) to protect against region-wide outages or vendor lock-in.

Strategy	Description	RTO	RPO	Cost
Backup & Restore	Data is backed up to cheap storage (e.g., Amazon S3). Infrastructure is rebuilt only after a disaster.	Hours to Days	High (last backup)	\$ (Low)
Pilot Light	A "minimal" version is always running (e.g., just the database). App servers are turned on and scaled up only when needed.	Minutes to Hours	Low	\$\$ (Medium)
Warm Standby	A scaled-down but fully functional copy of the environment is always running in another region.	Seconds to Minutes	Very Low	\$\$\$ (High)
Multi-Site (Active-Active)	Full-scale infrastructure runs in 2+ regions simultaneously,	Near Zero	Near Zero	\$\$\$\$ (Highest)

	both serving live traffic.			
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